

Physical changes to reduce flexible-shaft vibration

Gulo 3000 | Powertrain Dynamics Group | ME 495 | Experimental characterization and passive design

Prototype first: 165 mm distal flywheel + 4.50 mm shaft | 44% lower peak; 39% less switching motion

1 Problem and experimental approach

Motor switching excites twist between the flywheels.
Task: explain the vibration and find four $\geq 30\%$ solutions.

PHOTO TO TAKE — complete test rig

Take a clear front/oblique photo. Label the motor, motor-side flywheel, flexible shaft, distal flywheel and both speed sensors. Keep all components visible.

CAD SCREENSHOTS TO ADD — physical changes

Show the baseline assembly beside option 2: a 165 mm distal wheel and 4.50 mm shaft. Dimension both. Add a small section view of option 4: an 80 mm oil-damper rotor, 1 mm gaps on each face, and a casing fixed to the frame.

MEASURE

IDENTIFY

VALIDATE

COMPARE

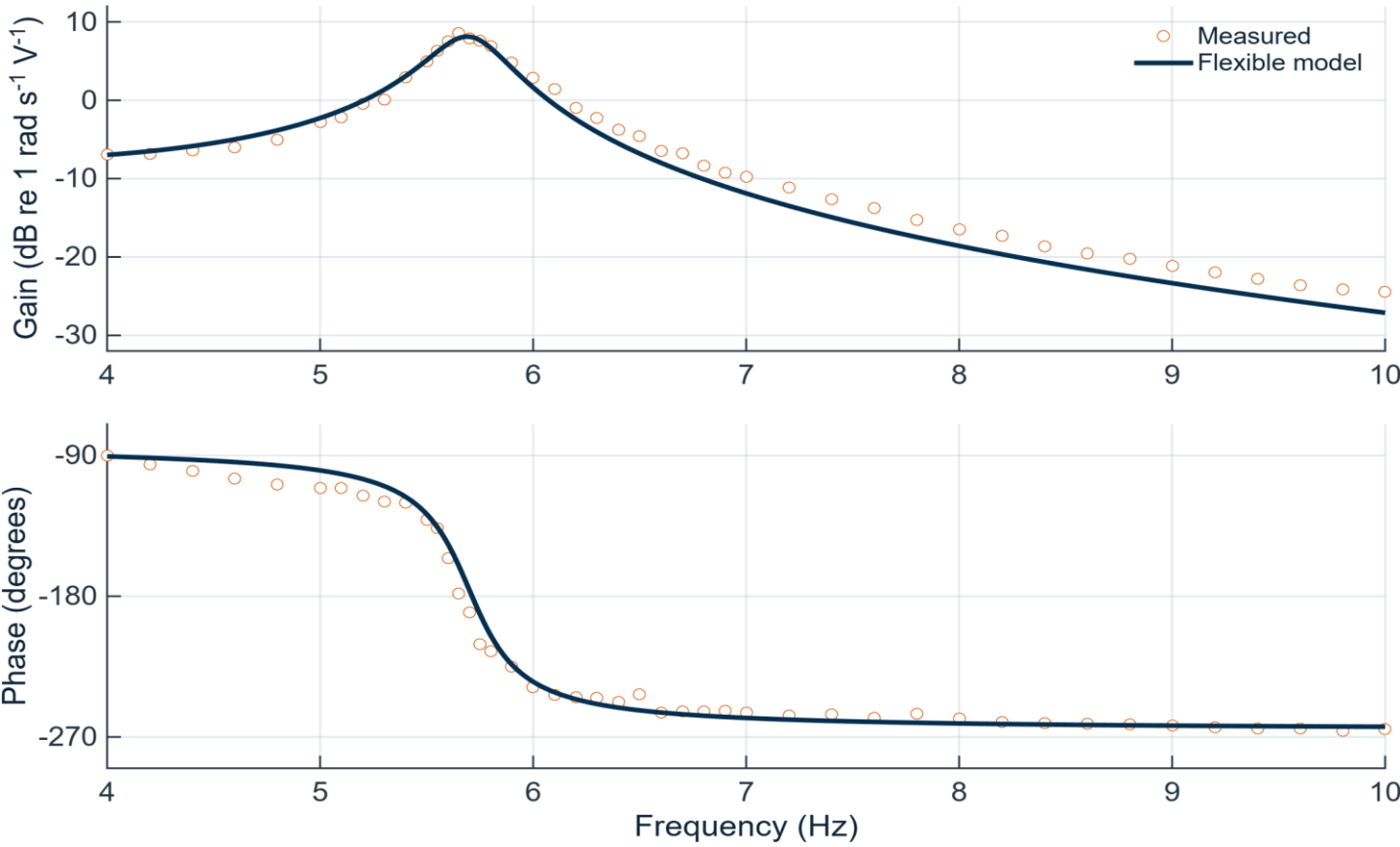
3 frequency sweeps + 28 time recordings

Steady runs identify the motor. Free-decay and coastdown tests estimate losses. Sweep data refine shaft stiffness and check the flexible model.

Measured / identified	Baseline value
Resonance	5.65 Hz
Motor resistance	2.495 Ω
Effective shaft stiffness	2.203 N·m/rad

Coastdown: motor physically uncoupled; both flywheels stayed linked by the shaft.

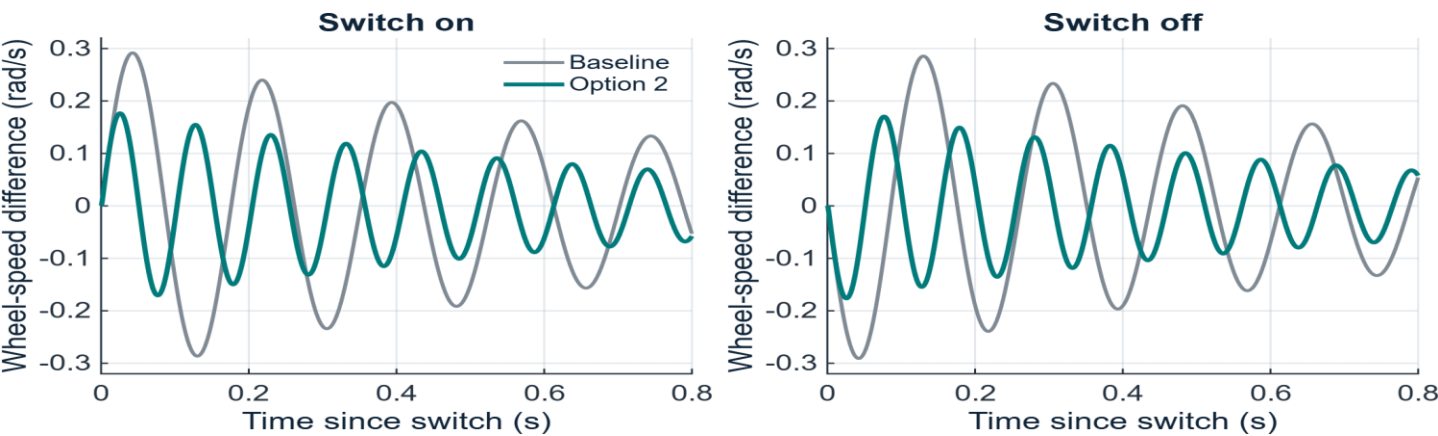
2 The flexible model matches the data



Validation metric	Measured	Model
Resonance frequency	5.65 Hz	5.69 Hz
Peak gain (rad/s/V)	2.69	2.55

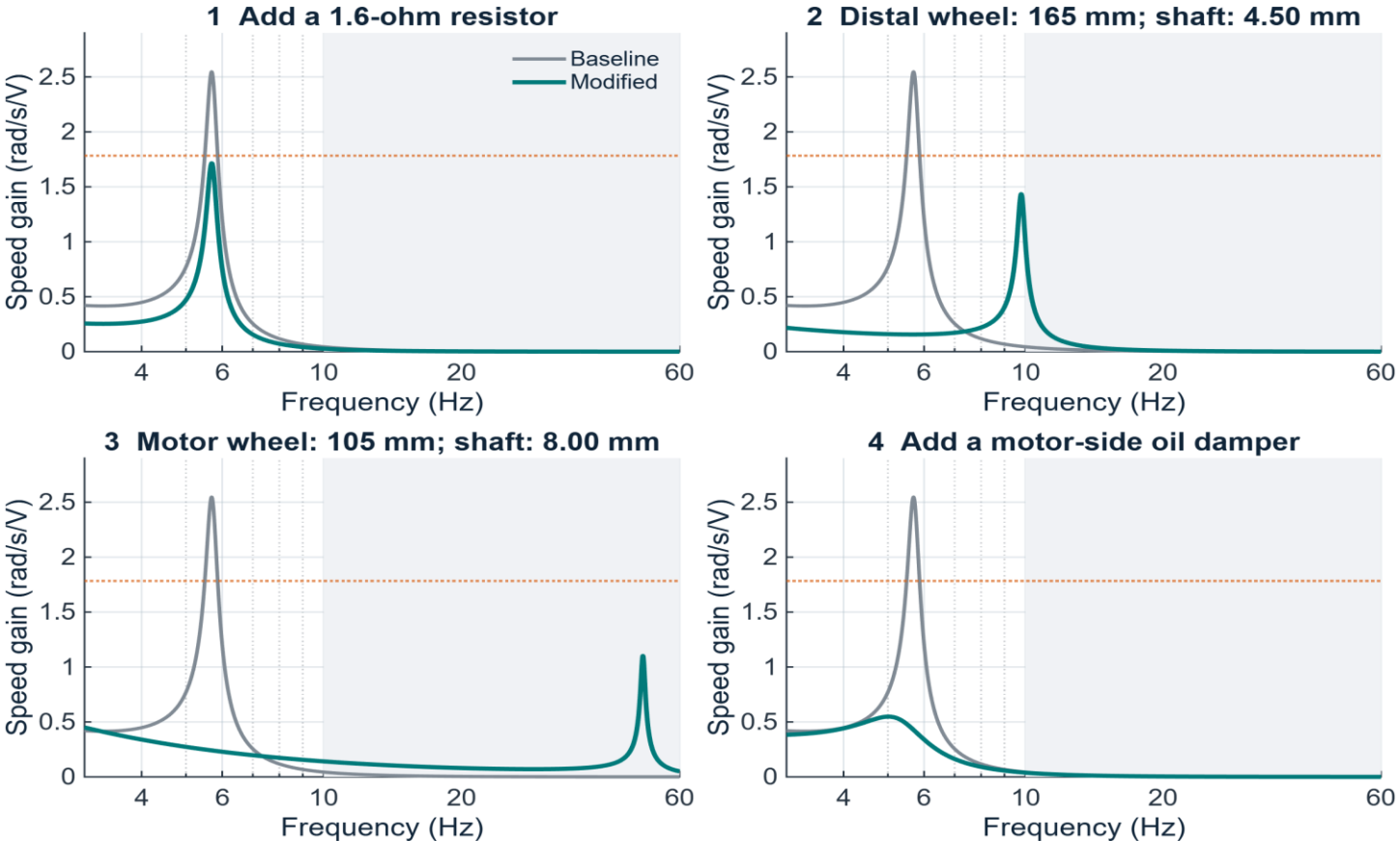
4–10 Hz fit error: 1.65 dB in gain; 7.02° in phase.

Option 2 also reduces switching motion



Simulation: +1 V for 4 s, then 0 V. Motion = difference between the two wheel speeds.

3 Four physical design alternatives



Dotted line: 30% peak reduction. Shading: beyond measured data.

Alternative	Peak ↓	Motion ↓
1 Series resistor	33%	38%
2 Larger distal wheel + shaft	44%	39%
3 Smaller motor wheel + shaft	57%	67%
4 Motor-side oil damper	78%	35%

All four also reduce the 1-Hz response cost by $\geq 30\%$.

- 1 Wire a 1.6 Ω power resistor in series; steady speed falls 29%.
- 2 Fit a 165 mm distal wheel + 4.50 mm shaft; preserves speed.
- 3 Fit a 105 mm motor wheel + 8.00 mm shaft; retest to 60 Hz.
- 4 Add an oil rotor in a fixed casing; steady speed falls 88%.

Full dimensions, mounting details and limitations: handout p. 3.